

ADITYA RAJ

Python Developer | Generative AI Engineer

adirajad2629@gmail.com · +91 9121855702 · Hyderabad, Telangana · linkedin.com/in/adityaraj2434 · github.com/adityaraj2434

PROFILE

Published researcher and 1st-Prize Hackathon winner with hands-on experience building production-grade Gen AI systems using LangChain, RAG pipelines, FAISS vector stores, and multi-agent workflows. Proficient in Python backend development, LLM API integration (OpenAI, Anthropic, Gemini), and prompt engineering. Seeking a fresher role in Generative AI or Python development where I can contribute to real-world AI products.

TECHNICAL SKILLS

AI / Gen AI

LangChain, LangGraph, RAG Pipelines, FAISS, Vector Stores, Prompt Engineering, AI Agents, Autonomous Workflows, LLM Fine-tuning basics

LLM APIs & SDKs

OpenAI API (GPT-4o), Anthropic Claude API, Google Gemini API, Hugging Face Transformers, Streamlit

Python Development

Python, FastAPI/Flask (basics), NumPy, Pandas, OpenCV, OOP, File Handling, Exception Handling

Web & Frontend

HTML, CSS, Basic JavaScript

Database

SQL (SELECT, WHERE, GROUP BY, JOINS), Basic Query Optimization

Tools & Platforms

Git, GitHub, VS Code, Jupyter Notebook, Google Colab

PROJECTS

Fusion Forge – Agentic AI Platform | *Python · LangChain · RAG · FAISS · Streamlit · LLM APIs*

- Designed and built a modular multi-agent AI platform integrating RAG pipeline with FAISS vector store for context-aware document Q&A across PDF, CSV, and plain-text inputs.
- Orchestrated autonomous agent workflows using LangChain with keyword-triggered real-time web data retrieval, reducing manual lookup time significantly.
- Engineered a sandboxed Python code execution engine enabling safe runtime evaluation inside the application — mirroring production-grade AI code assistant design.
- Built multilingual YouTube video summarization agents supporting English, Hindi, and Telugu using transcript extraction + LLM summarization chains.
- Implemented multi-turn conversational AI chatbot with persistent context management and prompt engineering for consistent, role-specific LLM behaviour.
- Research published as a book chapter in Taylor & Francis (2025) — DOI: 10.1201/9781003675242-35**

Agentic Voice AI Assistant [In Development] | *Python · OpenAI API · LangChain · Speech Recognition · TTS · Autonomous Agents*

- Designing a Jarvis-style autonomous voice assistant with a persistent agent persona — combining speech-to-text input, LLM reasoning (OpenAI API), and text-to-speech response into a seamless real-time loop.
- Architecting a LangChain-based tool-use agent capable of autonomously deciding when to search the web, execute code, retrieve documents, or answer from context — without explicit user commands.
- Engineering a character-consistent prompt layer using system-level prompt engineering to maintain personality, tone, and reasoning style across all interactions.
- Implementing a memory module for long-term conversational context retention across sessions — enabling the assistant to recall past interactions and user preferences.

Fake Currency Detection System | *Python · OpenCV · Machine Learning*

- Built an image processing pipeline using OpenCV for grayscale conversion, edge detection, and feature extraction to classify genuine vs. counterfeit currency notes.
- Applied basic ML classification logic achieving reliable detection on sample dataset — demonstrated ability to design end-to-end computer vision workflows.

EDUCATION

B.Tech – Computer Science Engineering KG Reddy College of Engineering & Technology, Hyderabad CGPA: 7.15 / 10.0	Oct 2021 – May 2025
Intermediate (CBSE) – 71%	2019 – 2021
Class X (CBSE) – 83%	2017 – 2019

AWARDS & ACHIEVEMENTS

- 1st Prize – 24-Hour Python Full-Stack Hackathon** · Brainovision, 2023
- Led a 4-member team from ideation to final presentation, winning against 20+ competing teams.
- Book Chapter Publication – Taylor & Francis (2025)**
- Co-authored peer-reviewed book chapter on AI tool integration as part of Multi-Disciplinary Research and Sustainable Development — DOI: 10.1201/9781003675242-35
- IELTS – C1 (Advanced)** · English Language Proficiency

LANGUAGES

English (IELTS C1 – Advanced) · **Hindi** (Native)